

Evaluating Functions from Rules, Tables, and Graphs

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

- | | |
|-----------------------------------|---|
| 1. 17 | 8. 7 |
| 2. 15 | 9. 2 |
| 3. 12 | 10. $s(1) = 5$, $s(4) = 14$, $s(4)$ minus $s(1) = 9$ |
| 4. 19 | 11. $m(2)$ from the graph = 7, $n(2)$ and the comparison symbol = = |
| 5. 48 | 12. the height $w(3) = 24$ cm, — |
| 6. < | |
| 7. the value of $f(-4) = -10$, — | |
-

What is verified

10 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 7, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.F.A – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

8.F.B – problem 12

Difficulty 1–5 of 5 across 17 tagged checks.

1. $f(x) = 4x - 3$. Find $f(5)$.

Solution: The input is 5, so replace every x in the rule with 5: $f(5) = 4(5) - 3 = 20 - 3$.

$f(5) = 17$

2. $g(x) = -2x + 9$. Find $g(-3)$.

Solution: Replace x with -3 and keep the negative sign attached to the input:
 $g(-3) = -2(-3) + 9 = 6 + 9$.

$g(-3) = 15$

3. $h(x) = x^2 - 4x$. Find $h(-2)$.

Solution: Substitute -2 for x in both terms. Square first, then multiply:

$$h(-2) = (-2)^2 - 4(-2) = 4 + 8$$

The squared term is positive and the subtracted term becomes an addition. $h(-2) = 12$

4. The table gives the function p . Find $p(3)$.

Solution: $p(3)$ asks for the output paired with the input 3. Find 3 in the x row and read straight down to the $p(x)$ row, which holds 19. $p(3) = 19$

5. The table gives $c(n)$, the cost in dollars of n tickets. Find $c(4)$.

Solution: The input is a number of tickets, so look for $n = 4$ in the top row. The cost paired with it is 48. (Check: each ticket costs 12, and $4 \times 12 = 48$.) $c(4) = 48$

6. Use $f(x) = 4x - 3$ and the table for p . Fill in $f(3)$, $p(3)$, and the comparison symbol.

Solution: Evaluate the rule at 3: $f(3) = 4(3) - 3 = 12 - 3 = 9$. Read the table at 3: $p(3) = 19$. Nine is to the left of nineteen on the number line, so the correct symbol is “less than”. $9 < 19$

7. $f(x) = 3x + 2$. Find $f(-4)$, then explain why $f(-4) \neq -f(4)$.

Solution: $f(-4) = 3(-4) + 2 = -12 + 2$. $f(-4) = -10$

Explanation (instructor-judged). A correct response says that $f(-4)$ puts the negative number *into* the rule, while $-f(4)$ evaluates $f(4) = 3(4) + 2 = 14$ first and then takes the opposite, giving -14 . The minus sign is applied at a different stage, so the two results differ: -10 against -14 . Accept any wording that makes that difference clear; give no credit for a numerical answer with no reason.

8. The graph of q is shown. Find $q(3)$.

Solution: Start at 3 on the horizontal axis, go straight up to the plotted point, then read across to the vertical axis. The point is $(3, 7)$. $q(3) = 7$

9. The graph of r is shown. For which input is the output of r greatest?

Solution: The plotted outputs are 4, 9, 6, 2 and 5. The highest point on the graph is $(2, 9)$, so the greatest output is 9 and it belongs to the input 2. The question asks for the input, not the output. $x = 2$

10. The graph of s is shown. Find $s(1)$, $s(4)$, and $s(4) - s(1)$.

Solution: Read the two plotted points: $(1, 5)$ gives $s(1) = 5$ and $(4, 14)$ gives $s(4) = 14$. Subtract the outputs in the order asked: $14 - 5 = 9$.

$s(1) = 5$	$s(4) = 14$	$s(4) - s(1) = 9$
------------	-------------	-------------------

11. The graph shows m ; the rule $n(x) = 4x - 1$ gives n . Compare $m(2)$ and $n(2)$.

Solution: From the graph, the point above 2 is $(2, 7)$, so $m(2) = 7$. From the rule, $n(2) = 4(2) - 1 = 8 - 1 = 7$. Two functions given in different forms can still agree at a particular input.

$m(2) = 7$	$n(2) = 7$	$m(2) = n(2)$
------------	------------	---------------

12. The graph gives w , the seedling's height in centimetres after x weeks. Find $w(3)$ and explain what it tells you.

Solution: The input 3 means three weeks after the first measurement. The plotted point there is $(3, 24)$, and the vertical axis is measured in centimetres.

$w(3) = 24 \text{ cm}$

Interpretation (instructor-judged). A correct sentence names both quantities with their units: three weeks after the first measurement the seedling was 24 centimetres tall. Accept any wording that ties the input to weeks and the output to a height; a bare repetition of the number earns no credit.